Alert Reader Study: full questionnaire

Every screen and question in the study app, in order. Participants see the 20 alerts in a shuffled order and with shuffled distractor options; this export uses one fixed order. The answer key is at the end (facilitators only).

Screen 1. Welcome (remote version)

Thank you for helping with our study. You have been given a random code below; just press Start (or type the code you were sent, if you have one). Please do it in one sitting of about 12 to 15 minutes, on a laptop or tablet if you can.

Participant code: _____ (your name is never asked for or stored)

Screen 2. Before you start

A monitoring system watches network traffic going to medical devices on a hospital network. When it decides a short window of traffic is an attack, it writes a note explaining why. You will read 20 of these notes and answer a few questions about each.

Imagine you are on night duty in a hospital's IT room. The monitoring system raises 20 alerts in rounds of five. For each one you decide why it fired, what you would do, and how much you trust it. No technical background is needed: we are testing whether the notes are understandable, not testing you. It takes about 12 to 15 minutes, and you see your score at the end.

Taking part is voluntary. You can stop at any point without giving a reason, and your answers are then discarded. Answers are stored under a code, never your name, and are reported only as group statistics.

- ☐ I have read this information, I am 18 or older, and I agree to take part. Ticking this box is my consent.

Screen 3. About you

Which best describes you?

- ○ Undergraduate student
- ○ Postgraduate student
- ○ Researcher or faculty
- ○ Clinician or nurse
- ○ IT or network staff
- ○ Other

Hands-on experience with computer networks

- ○ None
- ○ Less than 1 year
- ○ 1 to 3 years
- ○ More than 3 years

How familiar are you with machine learning?

1 (not at all) 2 3 4 5 6 7 (very)

How familiar are you with network security?

1 (not at all) 2 3 4 5 6 7 (very)

Screens 4 to 23. The 20 alerts (four rounds of five)

After alerts 5, 10 and 15 a short progress screen appears: "Round N of 4 complete, about X minutes to go. Your score is revealed at the end."

Alert 1 of 20 (Round 1 of 4)

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 100%, against 20% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       20% of the total evidence, strongest at record 7 of 10.
  5. the size of the smallest packet in the conversation (Pkt_Len_Min)
       6% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       9% of the total evidence, strongest at record 9 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       8% of the total evidence, strongest at record 6 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       6% of the total evidence, strongest at record 3 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 1.05s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average gap between replies from the device (Bwd_IAT_Mean)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 2 of 20 (Round 1 of 4)

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 92%, against 18% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       18% of the total evidence, strongest at record 8 of 10.

What argued against it (these looked more like Normal):
  2. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       12% of the total evidence, strongest at record 9 of 10.
  3. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       7% of the total evidence, strongest at record 8 of 10.
  4. the size of the largest packet in the conversation (Pkt_Len_Max)
       5% of the total evidence, strongest at record 9 of 10.
  5. how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
       5% of the total evidence, strongest at record 3 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.85s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the shortest gap between replies from the device (Bwd_IAT_Min)
- ☐ the longest quiet period in the conversation (Idle_Max)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 3 of 20 (Round 1 of 4)

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 98%, against 19% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 98% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       23% of the total evidence, strongest at record 9 of 10.
  5. the size of the smallest packet in the conversation (Pkt_Len_Min)
       7% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       14% of the total evidence, strongest at record 9 of 10.
  3. the average packet size in the conversation (Pkt_Len_Mean)
       8% of the total evidence, strongest at record 10 of 10.
  4. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       7% of the total evidence, strongest at record 10 of 10.

Detector agreement: the cnn detector carried the most weight (34%); weights across all detectors
were cnn 34%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.98s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the average gap between packets in the conversation (Flow_IAT_Mean)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 4 of 20 (Round 1 of 4)

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 98%, against 20% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       19% of the total evidence, strongest at record 7 of 10.

What argued against it (these looked more like Normal):
  2. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       14% of the total evidence, strongest at record 9 of 10.
  3. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       7% of the total evidence, strongest at record 9 of 10.
  4. the size of the largest packet in the conversation (Pkt_Len_Max)
       5% of the total evidence, strongest at record 7 of 10.
  5. how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
       5% of the total evidence, strongest at record 8 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.99s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ how much data the device said it was ready to receive when it first replied (Init_Bwd_Win_Byts)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 5 of 20 (Round 1 of 4)

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +0.0% at k=2,
threshold 5%) — the model's decision here is spread across many measurements, so no short list
explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the smallest packet sent to the device (Fwd_Pkt_Len_Min)
       13% of the total evidence, strongest at record 8 of 10.
  2. how much data the device said it was ready to receive when it first replied
(Init_Bwd_Win_Byts)
       11% of the total evidence, strongest at record 6 of 10.
  4. which network protocol was used (TCP, UDP, or other) (Protocol)
       9% of the total evidence, strongest at record 8 of 10.
  5. the total size of the addressing information on the device's replies (Bwd_Header_Len)
       6% of the total evidence, strongest at record 8 of 10.

What argued against it (these looked more like Normal):
  3. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       11% of the total evidence, strongest at record 2 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.81s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
- ☐ which network protocol was used (TCP, UDP, or other) (Protocol)
- ☐ the total size of the addressing information on the device's replies (Bwd_Header_Len)
- ☐ how much data the device said it was ready to receive when it first replied (Init_Bwd_Win_Byts)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the smallest packet sent to the device (Fwd_Pkt_Len_Min)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 6 of 20 (Round 2 of 4)

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 100%, against 20% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       22% of the total evidence, strongest at record 8 of 10.
  4. the size of the smallest packet in the conversation (Pkt_Len_Min)
       6% of the total evidence, strongest at record 8 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       11% of the total evidence, strongest at record 9 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       10% of the total evidence, strongest at record 9 of 10.
  5. the average packet size in the conversation (Pkt_Len_Mean)
       6% of the total evidence, strongest at record 4 of 10.

Detector agreement: the bilstm detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.87s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ the longest quiet period in the conversation (Idle_Max)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 7 of 20 (Round 2 of 4)

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +0.3% at k=3,
threshold 5%) — the model's decision here is spread across many measurements, so no short list
explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
       9% of the total evidence, strongest at record 8 of 10.
  3. the longest pause between packets sent to the device (Fwd_IAT_Max)
       8% of the total evidence, strongest at record 8 of 10.

What argued against it (these looked more like Normal):
  2. the average gap between replies from the device (Bwd_IAT_Mean)
       8% of the total evidence, strongest at record 1 of 10.
  4. the shortest gap between replies from the device (Bwd_IAT_Min)
       7% of the total evidence, strongest at record 1 of 10.
  5. the average gap between packets in the conversation (Flow_IAT_Mean)
       7% of the total evidence, strongest at record 2 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.85s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average gap between replies from the device (Bwd_IAT_Mean)
- ☐ the shortest gap between replies from the device (Bwd_IAT_Min)
- ☐ the average gap between packets in the conversation (Flow_IAT_Mean)
- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the longest pause between packets sent to the device (Fwd_IAT_Max)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 8 of 20 (Round 2 of 4)

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 99%, against 20% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 99% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       22% of the total evidence, strongest at record 6 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       11% of the total evidence, strongest at record 7 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       8% of the total evidence, strongest at record 7 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       7% of the total evidence, strongest at record 7 of 10.
  5. the size of the largest packet in the conversation (Pkt_Len_Max)
       6% of the total evidence, strongest at record 7 of 10.

Detector agreement: the cnn detector carried the most weight (34%); weights across all detectors
were cnn 34%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.74s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 9 of 20 (Round 2 of 4)

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain -0.0% at k=2,
threshold 5%) — the model's decision here is spread across many measurements, so no short list
explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  2. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       10% of the total evidence, strongest at record 9 of 10.
  4. the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
       6% of the total evidence, strongest at record 9 of 10.
  5. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       5% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       18% of the total evidence, strongest at record 1 of 10.
  3. the smallest packet sent to the device (Fwd_Pkt_Len_Min)
       8% of the total evidence, strongest at record 9 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.98s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the total size of the addressing information on the device's replies (Bwd_Header_Len)
- ☐ the smallest packet sent to the device (Fwd_Pkt_Len_Min)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 10 of 20 (Round 2 of 4)

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 99%, against 19% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 99% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       24% of the total evidence, strongest at record 9 of 10.
  5. the size of the smallest packet in the conversation (Pkt_Len_Min)
       6% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       12% of the total evidence, strongest at record 9 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       7% of the total evidence, strongest at record 2 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       7% of the total evidence, strongest at record 9 of 10.

Detector agreement: the cnn detector carried the most weight (34%); weights across all detectors
were cnn 34%, bilstm 33%, transformer 34%.
(Explanation produced by SHAP in 1.01s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ the total size of the addressing information on the device's replies (Bwd_Header_Len)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average quiet gap between bursts of activity (Idle_Mean)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 11 of 20 (Round 3 of 4)

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +0.1% at k=3,
threshold 5%) — the model's decision here is spread across many measurements, so no short list
explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       11% of the total evidence, strongest at record 4 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       7% of the total evidence, strongest at record 7 of 10.
  5. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       6% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  2. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       11% of the total evidence, strongest at record 2 of 10.
  3. the average size of packets in the conversation (Pkt_Size_Avg)
       8% of the total evidence, strongest at record 5 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 1.00s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the smallest packet sent to the device (Fwd_Pkt_Len_Min)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 12 of 20 (Round 3 of 4)

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain -0.0% at k=2,
threshold 5%) — the model's decision here is spread across many measurements, so no short list
explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. how much data the device said it was ready to receive when it first replied
(Init_Bwd_Win_Byts)
       13% of the total evidence, strongest at record 4 of 10.
  3. the total size of the addressing information on the device's replies (Bwd_Header_Len)
       7% of the total evidence, strongest at record 4 of 10.
  4. which network protocol was used (TCP, UDP, or other) (Protocol)
       7% of the total evidence, strongest at record 1 of 10.

What argued against it (these looked more like Normal):
  2. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       8% of the total evidence, strongest at record 4 of 10.
  5. how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
       5% of the total evidence, strongest at record 6 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.86s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ which network protocol was used (TCP, UDP, or other) (Protocol)
- ☐ how much data the device said it was ready to receive when it first replied (Init_Bwd_Win_Byts)
- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
- ☐ the total size of the addressing information on the device's replies (Bwd_Header_Len)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 13 of 20 (Round 3 of 4)

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +1.0% at k=5,
threshold 5%) — the model's decision here is spread across many measurements, so no short list
explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       13% of the total evidence, strongest at record 9 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       9% of the total evidence, strongest at record 10 of 10.
  5. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       7% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       10% of the total evidence, strongest at record 10 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       9% of the total evidence, strongest at record 10 of 10.

Detector agreement: the bilstm detector carried the most weight (33%); weights across all
detectors were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 1.03s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the average quiet gap between bursts of activity (Idle_Mean)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 14 of 20 (Round 3 of 4)

```
[VERIFIED] Removing the top 2 cited measurements changes this decision by 14%, against 0% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       9% of the total evidence, strongest at record 3 of 10.
  3. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       9% of the total evidence, strongest at record 9 of 10.
  5. the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
       6% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       14% of the total evidence, strongest at record 2 of 10.
  4. the smallest packet sent to the device (Fwd_Pkt_Len_Min)
       7% of the total evidence, strongest at record 9 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.96s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the smallest packet sent to the device (Fwd_Pkt_Len_Min)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 15 of 20 (Round 3 of 4)

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain -13.4% at
k=5, threshold 5%) — the model's decision here is spread across many measurements, so no short
list explains it

ALERT: this traffic window was classified as ATTACK with 68% confidence.

What pointed to Attack:
  3. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       7% of the total evidence, strongest at record 6 of 10.
  5. the average packet size in the conversation (Pkt_Len_Mean)
       5% of the total evidence, strongest at record 6 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       18% of the total evidence, strongest at record 2 of 10.
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       11% of the total evidence, strongest at record 9 of 10.
  4. which network protocol was used (TCP, UDP, or other) (Protocol)
       6% of the total evidence, strongest at record 1 of 10.

Detector agreement: the bilstm detector carried the most weight (39%); weights across all
detectors were cnn 32%, bilstm 39%, transformer 29%.
(Explanation produced by SHAP in 8.19s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
- ☐ which network protocol was used (TCP, UDP, or other) (Protocol)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ how long the connection lasted (Flow_Duration)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 16 of 20 (Round 4 of 4)

```
[VERIFIED] Removing the top 5 cited measurements changes this decision by 9%, against 0% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       12% of the total evidence, strongest at record 8 of 10.
  4. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       4% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       13% of the total evidence, strongest at record 2 of 10.
  3. the size of the largest packet in the conversation (Pkt_Len_Max)
       5% of the total evidence, strongest at record 8 of 10.
  5. how many packets were acknowledgements of data already received (ACK_Flag_Cnt)
       4% of the total evidence, strongest at record 2 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.93s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the shortest gap between replies from the device (Bwd_IAT_Min)
- ☐ how long the connection lasted (Flow_Duration)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ how many packets were acknowledgements of data already received (ACK_Flag_Cnt)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 17 of 20 (Round 4 of 4)

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +1.7% at k=3,
threshold 5%) — the model's decision here is spread across many measurements, so no short list
explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       8% of the total evidence, strongest at record 5 of 10.
  3. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       5% of the total evidence, strongest at record 9 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       19% of the total evidence, strongest at record 9 of 10.
  4. the smallest packet sent to the device (Fwd_Pkt_Len_Min)
       5% of the total evidence, strongest at record 9 of 10.
  5. the size of the largest packet in the conversation (Pkt_Len_Max)
       5% of the total evidence, strongest at record 3 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 1.08s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the smallest packet sent to the device (Fwd_Pkt_Len_Min)
- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 18 of 20 (Round 4 of 4)

```
[VERIFIED] Removing the top 2 cited measurements changes this decision by 7%, against 0% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       7% of the total evidence, strongest at record 8 of 10.
  3. the largest packet sent to the device (Fwd_Pkt_Len_Max)
       7% of the total evidence, strongest at record 9 of 10.
  5. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       5% of the total evidence, strongest at record 10 of 10.

What argued against it (these looked more like Normal):
  1. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       19% of the total evidence, strongest at record 9 of 10.
  4. the size of the largest packet in the conversation (Pkt_Len_Max)
       6% of the total evidence, strongest at record 8 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 0.95s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the size of the largest packet in the conversation (Pkt_Len_Max)
- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the largest packet sent to the device (Fwd_Pkt_Len_Max)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the longest quiet period in the conversation (Idle_Max)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 19 of 20 (Round 4 of 4)

```
[UNVERIFIED] removing the cited measurements barely changes this decision (best gain +0.0% at k=5,
threshold 5%) — the model's decision here is spread across many measurements, so no short list
explains it

ALERT: this traffic window was classified as ATTACK with 100% confidence.

What pointed to Attack:
  1. how long the conversation lasted (Flow_Duration)
       12% of the total evidence, strongest at record 10 of 10.
  3. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       8% of the total evidence, strongest at record 8 of 10.
  4. the longest quiet period in the conversation (Idle_Max)
       7% of the total evidence, strongest at record 10 of 10.
  5. the average packet size in the conversation (Pkt_Len_Mean)
       6% of the total evidence, strongest at record 8 of 10.

What argued against it (these looked more like Normal):
  2. the average size of packets in the conversation (Pkt_Size_Avg)
       10% of the total evidence, strongest at record 8 of 10.

Detector agreement: the cnn detector carried the most weight (33%); weights across all detectors
were cnn 33%, bilstm 33%, transformer 33%.
(Explanation produced by SHAP in 1.05s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the average time between packets sent to the device (Fwd_IAT_Mean)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ how long the conversation lasted (Flow_Duration)
- ☐ the average size of packets sent to the device (Fwd_Pkt_Len_Mean)
- ☐ the longest quiet period in the conversation (Idle_Max)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Alert 20 of 20 (Round 4 of 4)

```
[VERIFIED] Removing the top 1 cited measurement changes this decision by 99%, against 19% for
unrelated ones.

ALERT: this traffic window was classified as ATTACK with 99% confidence.

What pointed to Attack:
  1. the average size of packets in the conversation (Pkt_Size_Avg)
       18% of the total evidence, strongest at record 7 of 10.
  5. the size of the smallest packet in the conversation (Pkt_Len_Min)
       5% of the total evidence, strongest at record 6 of 10.

What argued against it (these looked more like Normal):
  2. the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
       9% of the total evidence, strongest at record 10 of 10.
  3. how many packets were requests to open a new connection (a flood of these is how port scans
and SYN attacks look) (SYN_Flag_Cnt)
       6% of the total evidence, strongest at record 2 of 10.
  4. the average packet size in the conversation (Pkt_Len_Mean)
       6% of the total evidence, strongest at record 9 of 10.

Detector agreement: the bilstm detector carried the most weight (34%); weights across all
detectors were cnn 33%, bilstm 34%, transformer 33%.
(Explanation produced by SHAP in 0.93s. 'Attack' is the positive class.)
```

1. Why did the system raise this alert? Tick what it based the decision on.

- ☐ the longest quiet period in the conversation (Idle_Max)
- ☐ how many packets were requests to open a new connection (a flood of these is how port scans and SYN attacks look) (SYN_Flag_Cnt)
- ☐ the average size of packets in the conversation (Pkt_Size_Avg)
- ☐ the size of the smallest packet in the conversation (Pkt_Len_Min)
- ☐ total time spent waiting between packets sent to the device (Fwd_IAT_Tot)
- ☐ the average amount of useful data per packet the device sent back (Bwd_Seg_Size_Avg)
- ☐ the average packet size in the conversation (Pkt_Len_Mean)

2. You're on duty. What do you do?

- ○ Act on it now
- ○ Ask someone to check first
- ○ Ignore it

3. How sure are you of your answer to 1?

1 (guessing) 2 3 4 5 6 7 (certain)

4. How much do you trust this alert?

1 (not at all) 2 3 4 5 6 7 (completely)

Add a comment (optional): _____

Screen 24. Last questions

Did any alerts look less reliable than the others? Which, and what told you?

_____

Was anything hard to understand? Quote any word or phrase that stopped you.

_____

The notes helped me understand why traffic was flagged.

1 (strongly disagree) 2 3 4 5 6 7 (strongly agree)

I could tell which notes the system was unsure about.

1 (strongly disagree) 2 3 4 5 6 7 (strongly agree)

I would rely on these notes in a real hospital network.

1 (strongly disagree) 2 3 4 5 6 7 (strongly agree)

Screen 25. Your results and debrief

Shows: reasons identified correctly (out of 20), unreliable alerts the participant did not act on blindly (out of 6), median time per alert.

Debrief: "Some of the notes you read were ones the system itself marks as unreliable, shown by the status line at the top. We did not tell you that in advance because we wanted to see whether the line is noticed on its own."

Then: "Save my answers" and send the file to the person who invited you.

Answer key (facilitators only)

Question 1 is correct if the reader ticks at least one measurement that pointed to Attack and ticks no counter-evidence and no distractor.

#	Type	Correct (pointed to Attack)	Wrong: counter-evidence	Wrong: distractors
1	Certified	Pkt_Size_Avg, Pkt_Len_Min	Bwd_Seg_Size_Avg, SYN_Flag_Cnt, Pkt_Len_Mean	Fwd_IAT_Mean, Bwd_IAT_Mean
2	Certified	Pkt_Size_Avg	SYN_Flag_Cnt, Bwd_Seg_Size_Avg, Pkt_Len_Max, ACK_Flag_Cnt	Bwd_IAT_Min, Idle_Max
3	Certified	Pkt_Size_Avg, Pkt_Len_Min	Bwd_Seg_Size_Avg, Pkt_Len_Mean, SYN_Flag_Cnt	Flow_IAT_Mean, Fwd_IAT_Mean
4	Certified	Pkt_Size_Avg	SYN_Flag_Cnt, Bwd_Seg_Size_Avg, Pkt_Len_Max, ACK_Flag_Cnt	Fwd_IAT_Tot, Init_Bwd_Win_Byts
5	Uncertified	Fwd_Pkt_Len_Min, Init_Bwd_Win_Byts, Protocol, Bwd_Header_Len	SYN_Flag_Cnt	ACK_Flag_Cnt, Pkt_Len_Max
6	Certified	Pkt_Size_Avg, Pkt_Len_Min	Bwd_Seg_Size_Avg, SYN_Flag_Cnt, Pkt_Len_Mean	ACK_Flag_Cnt, Idle_Max
7	Uncertified	Fwd_IAT_Tot, Fwd_IAT_Max	Bwd_IAT_Mean, Bwd_IAT_Min, Flow_IAT_Mean	Fwd_Pkt_Len_Max, Pkt_Size_Avg
8	Certified	Pkt_Size_Avg	Bwd_Seg_Size_Avg, SYN_Flag_Cnt, Pkt_Len_Mean, Pkt_Len_Max	Fwd_Pkt_Len_Mean, Fwd_IAT_Tot
9	Contradictory	Fwd_Pkt_Len_Max, Fwd_Pkt_Len_Mean, Bwd_Seg_Size_Avg	SYN_Flag_Cnt, Fwd_Pkt_Len_Min	Bwd_Header_Len, Fwd_IAT_Mean
10	Certified	Pkt_Size_Avg, Pkt_Len_Min	Bwd_Seg_Size_Avg, SYN_Flag_Cnt, Pkt_Len_Mean	Idle_Mean, Bwd_Header_Len
11	Uncertified	Bwd_Seg_Size_Avg, Pkt_Len_Mean, Fwd_Pkt_Len_Max	SYN_Flag_Cnt, Pkt_Size_Avg	Fwd_Pkt_Len_Min, Pkt_Len_Max
12	Uncertified	Init_Bwd_Win_Byts, Bwd_Header_Len, Protocol	SYN_Flag_Cnt, ACK_Flag_Cnt	Pkt_Size_Avg, Pkt_Len_Max
13	Uncertified	Bwd_Seg_Size_Avg, Pkt_Len_Mean, Fwd_Pkt_Len_Max	Pkt_Size_Avg, SYN_Flag_Cnt	Fwd_IAT_Mean, Idle_Mean
14	Contradictory	Pkt_Size_Avg, Fwd_Pkt_Len_Max, Fwd_Pkt_Len_Mean	SYN_Flag_Cnt, Fwd_Pkt_Len_Min	Pkt_Len_Min, Fwd_IAT_Tot
15	Contradictory	Bwd_Seg_Size_Avg, Pkt_Len_Mean	SYN_Flag_Cnt, Pkt_Size_Avg, Protocol	Flow_Duration, Fwd_IAT_Tot
16	Contradictory	Pkt_Size_Avg, Fwd_Pkt_Len_Max	SYN_Flag_Cnt, Pkt_Len_Max, ACK_Flag_Cnt	Bwd_IAT_Min, Flow_Duration
17	Contradictory	Pkt_Size_Avg, Fwd_Pkt_Len_Max	SYN_Flag_Cnt, Fwd_Pkt_Len_Min, Pkt_Len_Max	Fwd_Pkt_Len_Mean, Pkt_Len_Min
18	Contradictory	Pkt_Size_Avg, Fwd_Pkt_Len_Max, Bwd_Seg_Size_Avg	SYN_Flag_Cnt, Pkt_Len_Max	Fwd_IAT_Mean, Idle_Max
19	Uncertified	Flow_Duration, Bwd_Seg_Size_Avg, Idle_Max, Pkt_Len_Mean	Pkt_Size_Avg	Fwd_Pkt_Len_Mean, Fwd_IAT_Mean

20	Certified	Pkt_Size_Avg, Pkt_Len_Min	Bwd_Seg_Size_Avg, SYN_Flag_Cnt, Pkt_Len_Mean	Fwd_IAT_Tot, Idle_Max